

A Primitive-Direction Heat Transform for Square-Billiard Families

Route-A structural certificate C152

Abstract

We regularize the primitive direction families of the unit square billiard by a positive squared-length heat transform. Möbius inversion gives an exact theta factorization, and an elementary quarter-disk count plus Stieltjes integration gives its small-time law. Coincident lengths retain direction multiplicity. The construction is not a wave trace, isolated-orbit determinant, or target identity.

1 Transform and factorization

For ordered positive coprime pairs, coordinate swap retained and axes excluded, set

$$H_{\text{prim}}(t) = \sum_{\substack{m,n \geq 1 \\ (m,n)=1}} e^{-4t(m^2+n^2)}, \quad t > 0. \quad (1)$$

The exponent is $-tL_{m,n}^2$ for square-billiard length $L_{m,n} = 2\sqrt{m^2+n^2}$. With $\theta_+(u) = \sum_{k \geq 1} e^{-uk^2}$, the coprimality identity gives

$$H_{\text{prim}}(t) = \sum_{d \geq 1} \mu(d) \theta_+(4td^2)^2. \quad (2)$$

The interchange is absolute because monotonicity gives

$$\theta_+(4td^2) \leq \int_0^\infty e^{-4td^2x^2} dx = \frac{\sqrt{\pi}}{4d\sqrt{t}},$$

so the absolute transformed sum is bounded by $\frac{\pi}{16t} \sum_{d \geq 1} d^{-2}$. Each ordered primitive direction appears once, so different pairs of the same length add their multiplicities.

2 Primitive density and small time

For real $R \geq 0$, let $Q(R)$ count ordered positive lattice points in the quarter disk and let $N(R)$ count the coprime ones. Unit-square comparison gives

$$Q(R) = \frac{\pi R^2}{4} + O(R+1).$$

Möbius inversion then yields

$$N(R) = \sum_{d \leq R/\sqrt{2}} \mu(d) Q(R/d) = \frac{3R^2}{2\pi} + O(R \log R). \quad (3)$$

Stieltjes integration by parts gives

$$H_{\text{prim}}(t) = 8t \int_0^\infty r e^{-4tr^2} N(r) dr.$$

Substitution of (3) gives

$$H_{\text{prim}}(t) = \frac{3}{8\pi t} + O\left(t^{-1/2} \log(1/t)\right), \quad t \downarrow 0. \quad (4)$$

3 Exact validation and boundary

If b_s counts every ordered positive representation $s = a^2 + b^2$ and c_s the primitive ones, (2) is coefficientwise $c_s = \sum_{d^2|s} \mu(d) b_{s/d^2}$. Producer and independent checker agree for every $s \leq 20000$ and on Euclidean radius counts through 200; SymPy independently verifies initial coefficients and the leading integral.

This is a direction-label heat transform. It is not a clean wave trace, a Gutzwiller or isolated-orbit determinant, or the spectral heat trace of the Dirichlet Laplacian.

The tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION). We claim no target divisor, functional equation or counting law, arithmetic/local factor, Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.